Harbour porpoise *Phocoena phocoena* click detectability estimates depend on behaviour and detection-classification method



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More details

Aims

*Estimate a) **Probability of click detection** $\hat{P}$ with uncertainty, b) **Behavioural bias** in $\hat{P}$ for different detection-classification methods and c) **Drivers** of bias in $\hat{P}$*

Why?

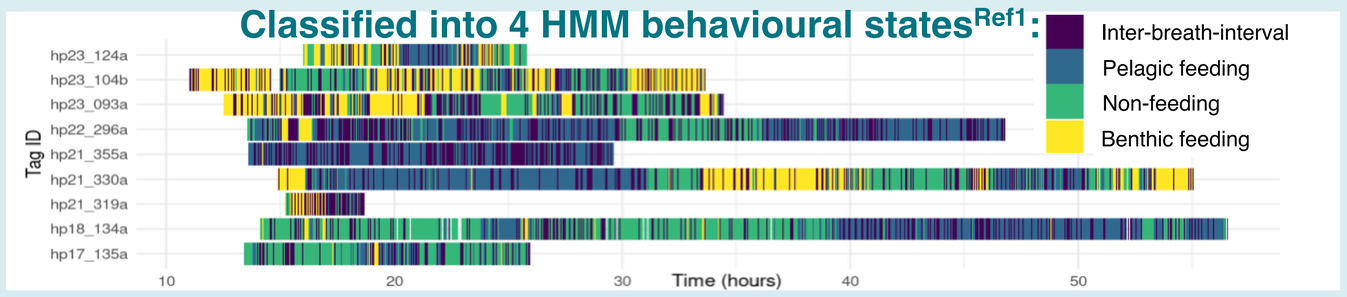
$\hat{P}$ is sensitive to porpoise echolocation behaviour, movement** and **detection-classification** methods, leading to **biased animal density estimates

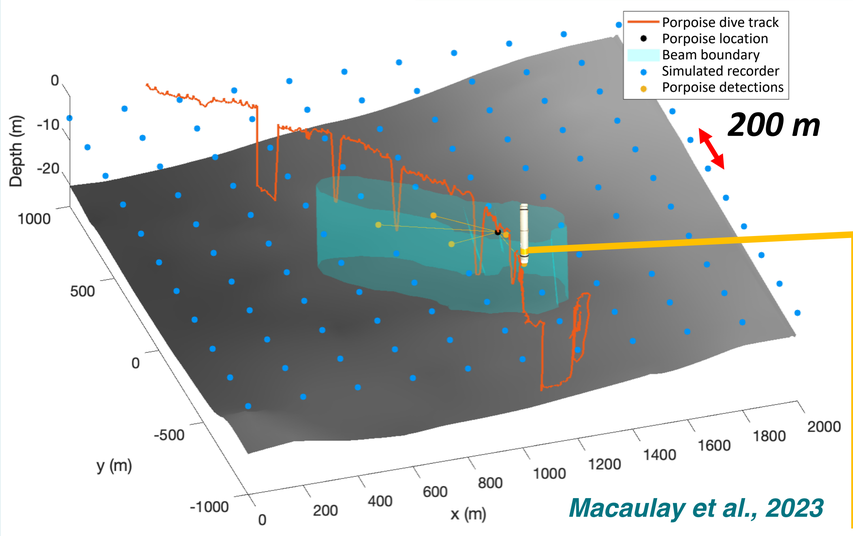
Methods



201hrs Harbour porpoise DTAG4 data

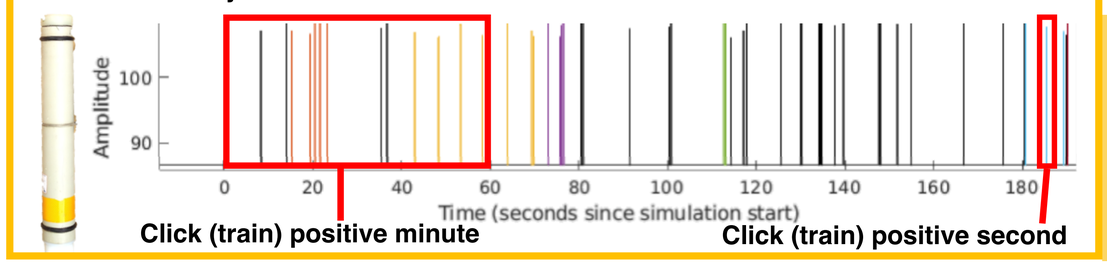
Classified into 4 HMM behavioural states[Ref1]:





Simulate CPOD detectors around 3D tag track, producing **>100,000 detection time series like this:**

Clicks **coloured by different click trains**. **Black non-train clicks deleted** for click-train-data

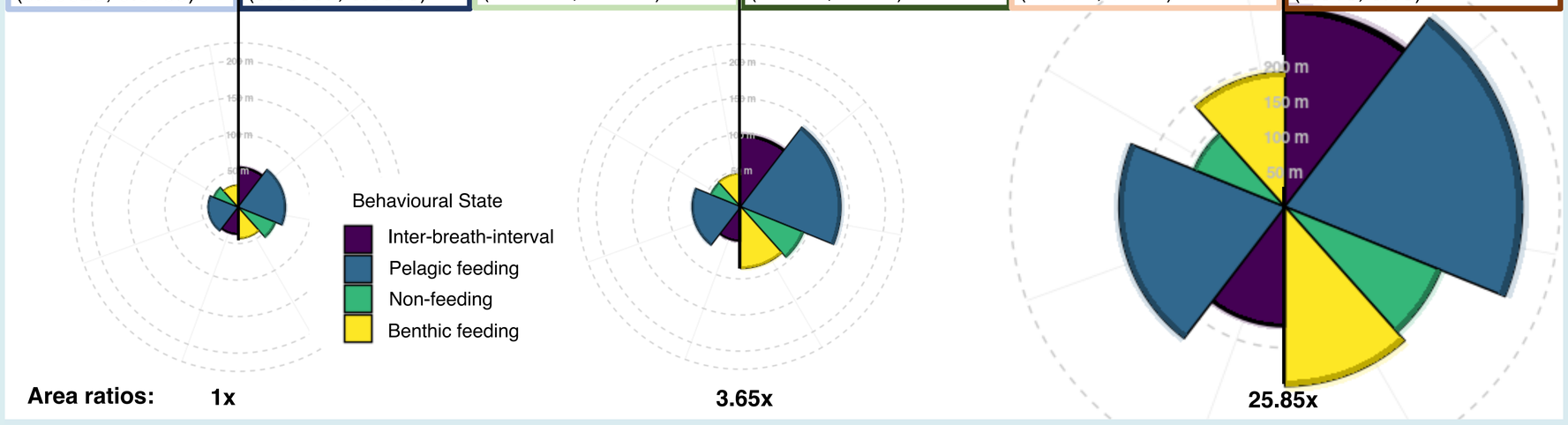


Custom **click train classifier** matching CPOD KERNO recall (~50%[Ref2, Ref3])

Results

a) Estimates of $\hat{P}$ and **effective detection area** around CPODs for 6 detection-classification methods by behaviour

Train click-by-click	Click-by-click	Train Positive Seconds	Click Positive Seconds	Train Positive Minutes	Click Positive Minutes
$\hat{P}$ = 0.0018 (0.00029, 0.0055)	$\hat{P}$ = 0.0035 (0.00067, 0.0096)	$\hat{P}$ = 0.0033 (0.00065, 0.0090)	$\hat{P}$ = 0.011 (0.0026, 0.028)	$\hat{P}$ = 0.031 (0.0081, 0.071)	$\hat{P}$ = 0.066 (0.018, 0.15)



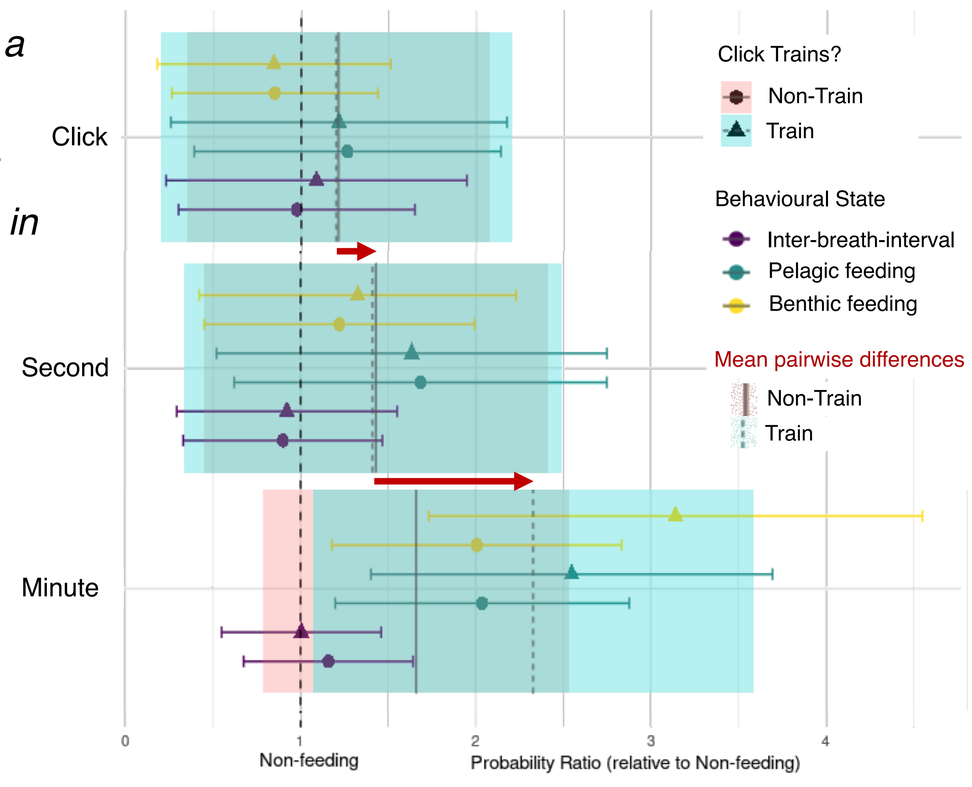
…6 dataset types:

	Click-by-click	Click-Positive-Seconds	Click-Positive-Minutes
No train filter	All clicks	Seconds with at least 1 click	Minutes with at least 1 click
Click train filtered	All clicks in click trains	Seconds with at least 1 click in click trains	Minutes with at least 1 click in click trains

b) Behavioural bias depending on detection-classification

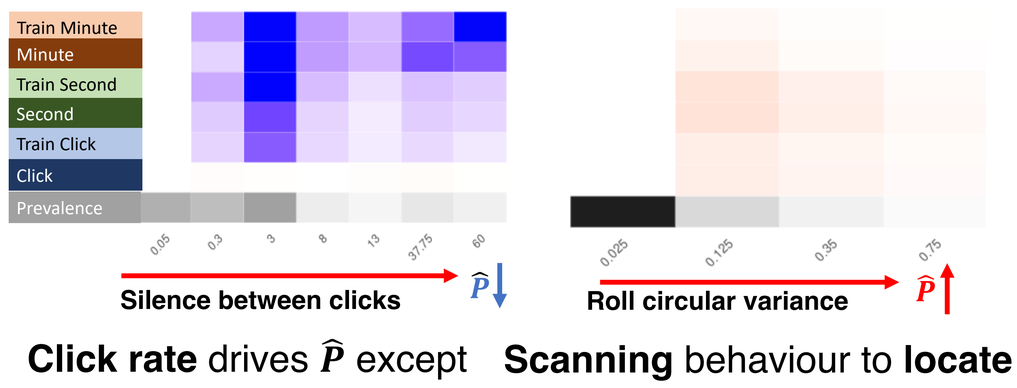
Predicted $\hat{P}$ from GLMM as a ratio of Non-feeding behavioural state (=1 in this plot) to illustrate differences in $\hat{P}$ between behaviours

Mean pairwise differences between the ratios for the 6 detection-classification units show **increasing $\hat{P}$ differences between behaviours from Click to Minute**



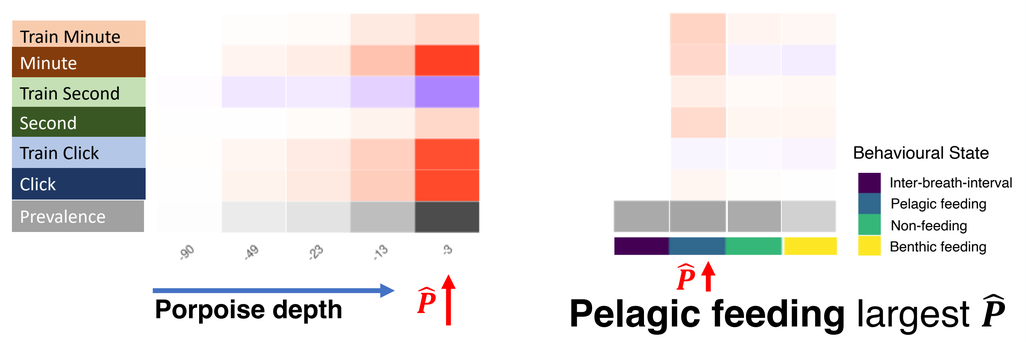
c) Behavioural drivers of $\hat{P}$

GLMM coefficients of binned behavioural variables weighted by prevalence (proportion of data spent in each bin) to estimate effect size (red = increased $\hat{P}$)



Click rate drives $\hat{P}$ except for click-by-click data

Scanning behaviour to **locate** and **track prey** drives $\hat{P}$



Dive behaviour interacts with **sensor depth** to drive $\hat{P}$

Pelagic feeding largest $\hat{P}$ increase for **positive seconds and minutes**

Conclusions and outlook

Behavioural states **differ most in $\hat{P}$** and **EDA** for **positive seconds** and **minutes,** click train classifier no improvement

Consistent with **previous work[Ref4]** we built on despite different tags/processing, georeferencing & detection time series

Implications: $\hat{P}$ behaviour sensitivity biases density estimates >2 fold. Problematic to transfer $\hat{P}$ between/within studies

Improve **detection-classification robustness,** estimate **behavioural state from monitoring data** to account for $\hat{P}$